R-Universe RFG-R

A regular multiscale completion with a scalar-action audit and empirical protocol

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Abstract. *RFG-R is a precisely specified low-energy completion of the R-Universe preferred-foliation branch. It regularizes the X=0 weak-field singularity while retaining the reconstructed cosmological background and luminal tensor propagation to O((epsilon/X)^p). The completion has two explicit domains: a cosmological relational action and a locally GR-matched weak-field EFT. The potential is reconstructed by quadrature from the prescribed branch rather than postulated. The local domain has gamma=beta=1 and alpha1=alpha2=0. The exact ADM-to-extended-EFT map contains a nonzero bar_m5 delta R3 delta K operator. An executed pure-gravity scalar audit retains that operator and finds a degenerate zeta-dot-squared coefficient, while the sourced photon--baryon--CDM--neutrino constraint action and untruncated kinetic hierarchy are explicitly derived. No RFG-R spectrum or CMB likelihood is evaluated. This paper defines the model, the audited limitation, and the empirical protocol; it makes no data-preference claim.*

Keywords: modified gravity; preferred foliation; CMB; post-Newtonian tests; effective field theory; standard sirens

1. The completion problem

The R-Universe homogeneous branch is predictive for H(a), the effective relational equation of state, and the tensor distance ratio. Its original response Q(X)=1-A X^(-theta), however, is singular in the X=0 limit used by a standard static weak-field expansion. A genuine empirical theory therefore needs a definition of its local limit as well as a derived linear matter and radiation system. RFG-R supplies the local completion and the action-level inputs for the latter. It is deliberately an effective low-energy model: cosmological geometry activates the relational response, while resolved local gravity is matched to General Relativity.

2. Regular cosmological response

Let p be even and larger than theta, with reference values theta=1.6, p=4, and epsilon=10^-8. Define nu=1+theta/p and A=Omega_R0/(1+theta).

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R_epsilon(X) = Omega_R0 X^(p+2)/(X^p+epsilon^p)^nu
Q_epsilon(X) = 1 - A X^p/(X^p+epsilon^p)^nu
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The covariant preferred-foliation action is the original RFG action after replacing Q and V by Q_epsilon and V_epsilon. Matter remains minimally and universally coupled to the Jordan metric.

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S_cos = (M_Pl^2/2) int d^4x sqrt(-g)
        { Q_epsilon(X)[R3 + K_mn K^mn - K^2] + 2 H0^2 V_epsilon(X) }
        + S_m[g,Psi]
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The FLRW lapse equation is fixed to be

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E^2(a) = Omega_m0 a^-3 + Omega_r0 a^-4 + R_epsilon(E)
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and determines the potential without a new function:

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F_epsilon = X^2 - R_epsilon - X^2 Q_epsilon - X^3 dQ_epsilon/dX
V_epsilon(X) = -3 X int_0^X ds F_epsilon(s)/s^2
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The integral is finite because F_epsilon/s^2 is O(X^p) at the origin. A term proportional to X is an ADM boundary representative and has no field-equation content.

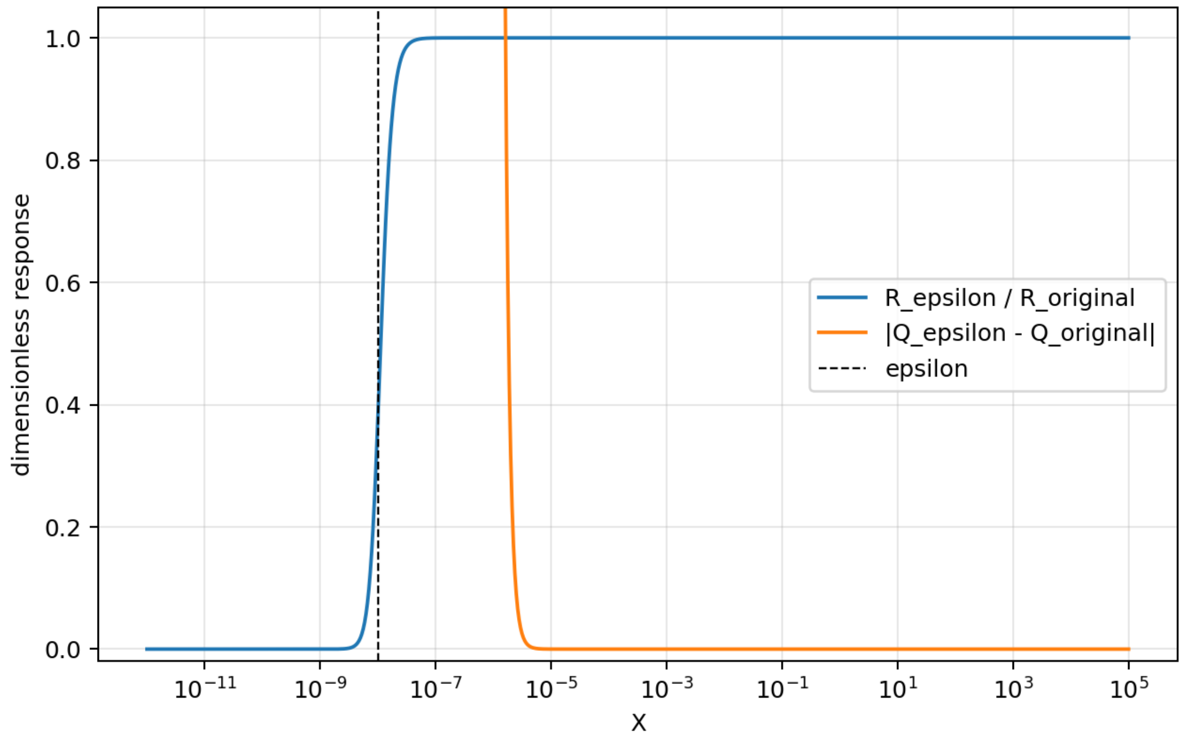

Figure 1. The regularized response becomes the original response immediately above the epsilon transition. On the cosmological branch the analytic difference is O((epsilon/X)^4).

3. Exact limits

At X much larger than epsilon the regularized branch recovers the original one. With the reference epsilon, the analytic relative correction is below 2.5e-32 for X>=0.8; double precision evaluates the equality at about 1e-15. At X=0 the action is regular.

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Q_epsilon = 1 - A epsilon^(-p-theta) X^p + O(X^(2p))
V_epsilon = -[3 Omega_R0(p-theta)/((1+theta)(p+1)epsilon^(p+theta))] X^(p+2)
              + O(X^(2p+2))
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For p=4 there is no physical correction to the quadratic Einstein-Hilbert ADM operator. This removes the X=0 singularity without altering the background or tensor branch that motivated RFG.

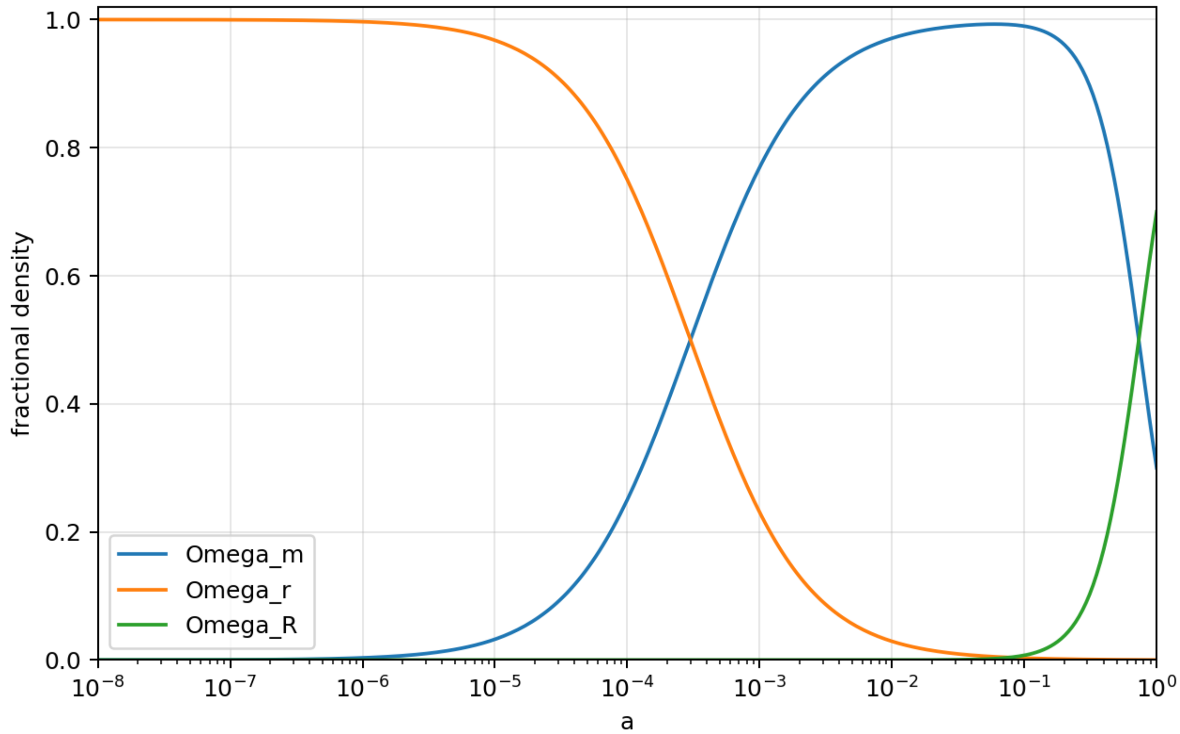

Figure 2. Fractional densities of the reference RFG-R branch. The regularization is absent at visible plotting precision on this cosmological domain.

4. Local GR matching and PPN

A single cosmological EFT is not automatically a local Solar-System EFT. RFG-R makes the crossover a model axiom. Define W=sqrt(abs(C_abcd C^abcd))/(H0/c)^2 and use a C-infinity switching function s(W) that equals one for W<=10^8 and zero for W>=10^9. The complete low-energy action is S_eff=S_GR+int s(W)[L_cos-L_GR]. FLRW has W=0, while a Schwarzschild Sun gives W=5.756e22 at one AU.

Inside the local matched domain s=0, so the action and all its variations are exactly those of GR. The PPN prediction is therefore gamma=1, beta=1, alpha1=0, and alpha2=0. The Cassini factor used by the likelihood is -2 ln L=[(gamma-1-2.1e-5)/(2.3e-5)]^2, yielding 0.833648 for the GR prediction.

5. Matter/CMB pre-Boltzmann gate and likelihood

No compressed CMB distance prior is used. The CMB interface begins with the action-level ADM derivatives. For L=Q_epsilon(X)[R3+K_ij K^ij-K^2]+2H0^2 V_epsilon(X), the principal inputs include L_R3=Q_epsilon and L_Kij_R3=Q_epsilon,X gamma^ij/(3H0). The exact ADM-to-extended-EFT map contains bar_m5=-M_Pl^2 Q_X/(3H0), multiplying bar_m5 delta R3 delta K/2. The repository generates all background functions and derivatives in eft_coefficients.csv, avoiding any differentiation of a singular power law.

Pure-gravity scalar audit

The mapped bar_m5 operator is retained in the reduced unitary-gauge scalar action. On a 49 by 49 logarithmic grid spanning 10^-7<=a<=1 and 10^-4<=k/H0<=10^5, the lapse--shift constraint discriminant remains positive but the zeta-dot-squared coefficient is degenerate at all 2,401 points; the maximum relative residual of its factorized numerator is 4.357e-16. A standalone scalar sound speed is therefore undefined. This blocks a direct one-scalar EFT/Boltzmann evolution. It does not itself prove that the physical matter-coupled theory is ill-posed, since matter changes the constraint system.

Canonical-scalar check

The sourced scalar constraints have been reduced exactly for one minimally coupled canonical field in comoving gauge. If m=phi_dot^2/M_Pl^2 and D=-F_XX/3, the lapse--shift determinant is proportional to Delta=6 D H^2 Q+(D-2Q)m and the reduced kinetic coefficient is K=6 D Q m/Delta. The intrinsic-curvature contribution +2Q(k^2/a^2)zeta^2 is retained; it is required to recover the exact GR massless-scalar result K=6 and c_s^2=1. On a self-consistent RFG-R reference branch with a massless canonical field, the executable diagnostic finds positive Delta, K, and gradient coefficients for 0.03<=a<=1. This is a real sourced-constraint check, not a dust/radiation/CMB calculation.

The sourced photon--baryon--CDM--neutrino finite quadratic constraint action is now reduced by an exact Schur complement. Its GR check has rank(K)=4 in rational arithmetic, and its 425-point RFG-R reference audit has inertia (positive, negative, null)=(4,0,1). Photon polarization and neutrino anisotropic stress remain untruncated kinetic hierarchies; massive neutrinos retain epsilon=sqrt(q^2+a^2m_nu^2). The remaining task is an action-faithful 3+1 Einstein-Boltzmann implementation with recombination and lensing. It must reject any point with a singular physical constraint matrix, a ghost, a gradient instability, or Q_T<=0. Accepted points would export TT, TE, EE, lensing, P(k,z), f sigma8(z), and the standard-siren distance.

6. Data protocol

The baseline log likelihood is ln L=ln L_Planck18+ln L_lensing+ln L_BAO+ln L_SN+ln L_RSD+ln L_PPN+ln L_GW. Planck is to be evaluated with the official low-ell temperature and polarization, high-ell TT/TE/EE, and lensing packages; the solver must be a full modified-gravity Einstein-Boltzmann solver rather than a quasi-static growth approximation. The public stock H-EFTCAMB interface is a useful GR reference, but it does not expose bar_m5 delta R3 delta K and is not an exact RFG-R solver. Cobaya may sample the joint likelihood only after an action-faithful extended solver, including recombination and lensing, is implemented and validated.

The completion defines a falsifiable route, but no RFG-R spectrum or data likelihood has yet been evaluated. Whether the model is preferred, allowed, or ruled out remains an empirical question; no conclusion follows from background algebra, local matching, a finite-core audit, or the pure-gravity audit alone.

7. Numerical verification

check	reference result
high-X response recovery	5.329e-15 numerical maximum
high-X Q recovery	2.600e-15 numerical maximum
potential reconstruction residual	8.145e-06 finite-difference maximum
background density closure	1.554e-13 maximum error
minimum Q_T over a in [1e-8,1e2]	0.6153847956
canonical scalar GR limit	K=6, c_s^2=1 exactly
canonical reference grid	Delta>0, K>0, G1>0, G2>=0 within numerical tolerance
pure-gravity extended-EFT scalar audit	2,401/2,401 degenerate; standalone c_s undefined
multi-fluid finite core	GR rank(K)=4 exactly; RFG-R inertia (4,0,1) on 425 points
Solar W at 1 AU	5.756e22; local GR domain

The checks are generated by scripts/run_all.sh. They test the action-defined completion and reproduce its pure-gravity scalar gate; they do not constitute a CMB or matter likelihood.

References

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